



Assignment 09: Working with Modules in Python



Instructions

1. Write each program in a separate Python file.
 2. Use Python's built-in modules wherever required.
 3. Display clear output messages.
 4. Import only the required modules.
 5. Test your programs before submission.
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1 Using the `math` Module

Problem:

Write a Python program that imports the `math` module and displays:

- Square root of 81
- Value of π (pi)
- Value of 5^2

Example Output

```
Square Root of 81 = 9.0  
Value of Pi = 3.141592653589793  
52 = 25.0
```



2 Generate a Random Number

Problem:

Write a Python program using the `random` module to generate a random number between 1 and 100.

Example Output

```
Random Number = 57
```

(Output will be different each time.)



3 Random Choice from a List

Problem:

Create a list of colors and display one random color using the `random.choice()` function.

Example Output

```
Selected Color = Blue
```

**4**

Display Current Date and Time

Problem:

Write a Python program using the `datetime` module to display the current date and time.

Example Output

```
Current Date and Time:  
2025-08-21 10:45:32
```

**5**

Display Current Year

Problem:

Write a Python program that displays only the current year using the `datetime` module.

Example Output

```
Current Year = 2025
```

**6**

Find the Current Working Directory

Problem:

Write a Python program using the `os` module to display the current working directory.

Example Output

```
Current Directory:  
C:\PythonPrograms
```

**7**

Create Your Own Module

Problem:

Create a file named `calculator.py` containing the following functions:

- `add(a, b)`
- `subtract(a, b)`

Then create another Python file that imports `calculator` and uses both functions.

Example Output

```
Addition = 30
Subtraction = 10
```



8 Import Specific Functions from a Module

Problem:

Import only the `sqrt()` function from the `math` module and calculate the square root of 144.

Example Output

```
Square Root = 12.0
```



Bonus Question

Student Utility Module

Step 1: Create a Module

Create a file named `student.py` containing:

```
def display_name(name):
    print("Student Name:", name)

def display_marks(marks):
    print("Marks:", marks)
```

Step 2: Use the Module

Import the module and call both functions.

Example Output

Student Name: Ahmed
Marks: 85

Learning Outcomes

After completing this assignment, students will be able to:

- ✓ Import built-in Python modules
- ✓ Use functions from the `math`, `random`, `datetime`, and `os` modules
- ✓ Import specific functions from modules
- ✓ Create custom modules
- ✓ Use modules in multiple Python files
- ✓ Organize programs into reusable components